

Q1. (10 marks) Explain why the surface $z = x^2 \cos y$ is orientable.

Q2. (20 marks) Compute the outward flux $\int_S F \cdot dS$, where S is the sphere $x^2 + y^2 + z^2 = 4$ and

$$F(x, y, z) = (x^2 + y^2 + z^2)(xi + yj + zk).$$

Q3. (20 marks) Compute $\int_C F \cdot dr$, where

$$F = \langle (x + y)^2, (x + y)^2, yz^3 \rangle,$$

and C is the curve formed by the intersection of $z = \sqrt{x^2 + y^2}$ and $(x - 1)^2 + y^2 = 1$ oriented in the counterclockwise direction when viewed from high above the xy -plane.

Q4. (20 marks) Prove Stokes' theorem for the special case where the surface is the graph of a function $z = f(x, y)$.

Q5. (10 marks) Define $f : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ by

$$f(x, y) = \begin{cases} 0 & \text{if } 0 \leq x < \frac{1}{2} \\ 1 & \text{if } \frac{1}{2} \leq x \leq 1. \end{cases}$$

Prove that f is integrable. Also, compute

$$\int_{[0,1] \times [0,1]} f.$$

Q6. (20 marks) Compute the value of

$$\int_{x^2 + y^2 + z^2 \leq 1} (x^2 + y^2 + z^2) dV.$$